

Elevator Pitch

Time Tower Defense: A desperate time traveller must fortify an abandoned medieval tower at the heart of the town and gather resources to repair their damaged time machine while defending against waves of interdimensional enemies known as the Swarm. Race against time to make it back to your time. How long can you last? How far can you make it before you're overrun?

Summary

Time Tower Defense is a strategic survival game that combines classic tower defense mechanics with resource management in a unique time travel setting. This game stands out by blending the strategic depth of tower defense with the added complexity of gathering and managing resources, all within a rich narrative framework. Players take on the role of the Traveller; he is a time-displaced hero who has crash-landed in a medieval city after sustaining heavy damage from the Swarm: mysterious, interdimensional enemies encountered during a dangerous journey through time. The core gameplay revolves around defending a fortified, abandoned tower in the heart of a medieval city. Simultaneously, the player must venture out to collect scattered resources and vital machine parts needed to repair the Traveller's damaged time machine. This dual-pressure system keeps players constantly engaged, as they must make critical decisions about when to strengthen their defenses and when to risk leaving the safety of the tower to gather what they need.

The game features a distinctive low-poly art style that brings the medieval cityscape to life with clean, geometric visuals, ensuring smooth performance across various hardware platforms and giving the game a unique visual identity that stands out in the competitive tower defense market. As relentless waves of Swarm creatures attack the tower through multiple access points, including drawbridges and ramparts, players must carefully plan and place their defenses while managing limited resources and time. Each round of defense is tense and rewarding, pushing players to balance their immediate survival needs with the long-term objective of repairing the time machine and returning home. The gameplay is designed for short, engaging sessions lasting 5-10 minutes, making it easy to pick up and play while still offering plenty of strategic depth. Every decision matters, from resource allocation to defensive placement, ensuring that players remain invested in both their short-term survival and their ultimate goal of escape.

Unique Selling Points

In a saturated market like tower defense games, we knew that if we wanted to break away from the crowd, it was crucial to create an innovative and modern version of this classic game. With that being said, we set out on a journey to put a new spin on a tower defense game by first incorporating time travel, building a destructible environment, and lastly, developing two modes of game play.

i) Time Travel Aesthetics

Most tower defense games rely on a single theme throughout the game. For example, Ancient Planet Tower Defense, created by the developer Moonlight Mouse, relies on the theme that an ancient civilization has been invaded by hordes of aliens. Bloon TD allows users to select a landscape at the beginning of the game (field, icescape, etc.), but remains consistent throughout and uses monkeys as the base of its theme. Other tower defense games are futuristic or WWII-esque, but overall they rely on a singular, predominant aesthetic. Our game, on the other hand, captures the interest of the user by incorporating a time travel element. Each level is set in a different time period and city to keep the game varied and the player captivated. This time travel aesthetic continues into the game, as instead of defending a "home base", the player will be defending each new city they have crash landed on, and their time machine which they are trying to repair in order to get back to their hometown.

ii) Destructible Environment

The mechanics of environment interaction play a key role in any tower defense game. The player relies on this interaction to stimulate interest and a desire to continue playing. Our game aims at increasing the level of interaction to capture the user's attention. One way in which we accomplish this is by incorporating destructible environment elements. The path to the time machine includes multiple routes that the enemy can take to attack the player. Along these routes, though, the user is able to place a variety of obstacles or barricades that the enemies will need to break down in order to proceed on that route. Additionally, other routes contain elements like bridges or buildings in the city that the character can destroy in order to force the enemies down a different, perhaps longer path. Unlike some tower defense games, where the player often has to sit back and wait during the enemy swarms to see if their defenses hold, this destructible environment keeps the game moving and more directly involves the user at every point in the game.

iii) Two Modes of Gameplay

While the majority of tower defense games focus on a singular goal, namely, to defend the home base, our game is multifaceted. There is, of course, the traditional gameplay which involves defending the time machine against swarms of enemies. However, our game also aims to incorporate a second mode of gameplay which includes a controllable character. This character needs to search for parts to repair their time machine in between those waves of enemies. This addition not only grants users more control in the game, but it is also more interactive and provides an element not really seen in this market before. The gameplay would work like this: the player sets down obstacles and defenses to block the enemies before the swarms, then after a wave has passed, they have a brief window of time to gather parts and resources for fixing the time machine. The enemies will progressively build up during the level and the swarms will get bigger so that if the character isn't focusing enough attention on fixing their Time Machine they will get overrun by them. A level of strategy is thus introduced to the game as well. Once again, this separates us from typical tower defense games and helps to further keep the player invested in the game.

The Team

Our team consists of four group members, and each of us will take on various roles throughout the development process. At the beginning of this project, we took the time to discuss our strengths and weaknesses and used this to subsequently shape our roles in the team. Since the majority of us are new to Unity and Game Development, it will be a collaborative effort to create a functional game in the scope that we have imagined. The team roles are as follows:

- I. Sumaylen (Developer/Programmer)**
 - A.** Take the lead in developing the key mechanics of the game
 - B.** Use experience with Blender to make 3D-models if necessary (could be helpful especially with the destructible environment elements)
 - C.** Focus on enemy swarm mode of gameplay (attacks and defenses and how the user interacts with them)
- II. Rebekah (Designer/Programmer)**
 - A.** Assist with programming elements (key mechanics and gameplay)
 - B.** Focus on resource collection gameplay mode (character movement, how the player uncovers resources, repairing the time machine)
 - C.** Assist with art style/level design
- III. Tade - (Designer/Audio Specialist)**
 - A.** Design and create main characters of the game (enemies, user character),
 - B.** Design sets (environments for each level theme - medieval, western, sci-fi, etc.)
 - C.** Tailor audio effects and background music in accordance with the level theme
- IV. Alex (Designer/Narrator)**

- A. Design maps/route for the enemies to take for each level design
- B. Create the “story arc” of the game (narration elements)
- C. Design and write the game tutorial and help screen instructions (player controls and how-tos for collecting resources)

The team will work cooperatively as joint project managers in order to stay on track with the production schedule. This will involve weekly group meetings (a continuation from the planning stage of game development) and discussion as needed via Slack.

Evidence of Prototyping

The first iteration of prototyping pulled ideas from various existing tower defense games. We knew early on that we wanted to begin with a medieval theme as the first level of the game. We created a very basic map layout in a low-fidelity wireframe prototype whilst keeping that theme in mind. The white squares shown in Figure 1 (below) serve as a rough example of what an enemy path to the time machine (represented by the car in Figure 1) could look like. We also included a bridge as part of the path which the user could then destroy to force the enemies back down the other route. Additionally, the legend in the bottom left hand corner was a first draft of the “toolbox” for the user - equipped with a variety of attacks available to them as well as defenses such as barricades that they could drag and drop into the game. This prototype was discussed during one of our group meetings, and we came to the conclusion that a few modifications should be made. For instance, in order to better incorporate the medieval theme, we decided that the time machine should be placed inside a castle tower at the center of the game environment. Barricades may include moats or mountains in line with the theme, and enemy paths would extend from all sides of the castle. At this point in the planning process we also decided to incorporate the two modes of gameplay. Thus, in the next iteration of the prototype, we wanted to include an idea of what the resource-collection mode of gameplay might look like. We settled on having two resources that the player would need to collect in order to fix the time machine - the first, a mineral for fuel that they would need to dig for in the city, and the second, lost parts from the time machine when it crash landed. We also decided to include a help icon which the player could click on to pull up controls and game instructions. These modifications were added to a second prototype in the form of a higher-fidelity hand-drawn wireframe.

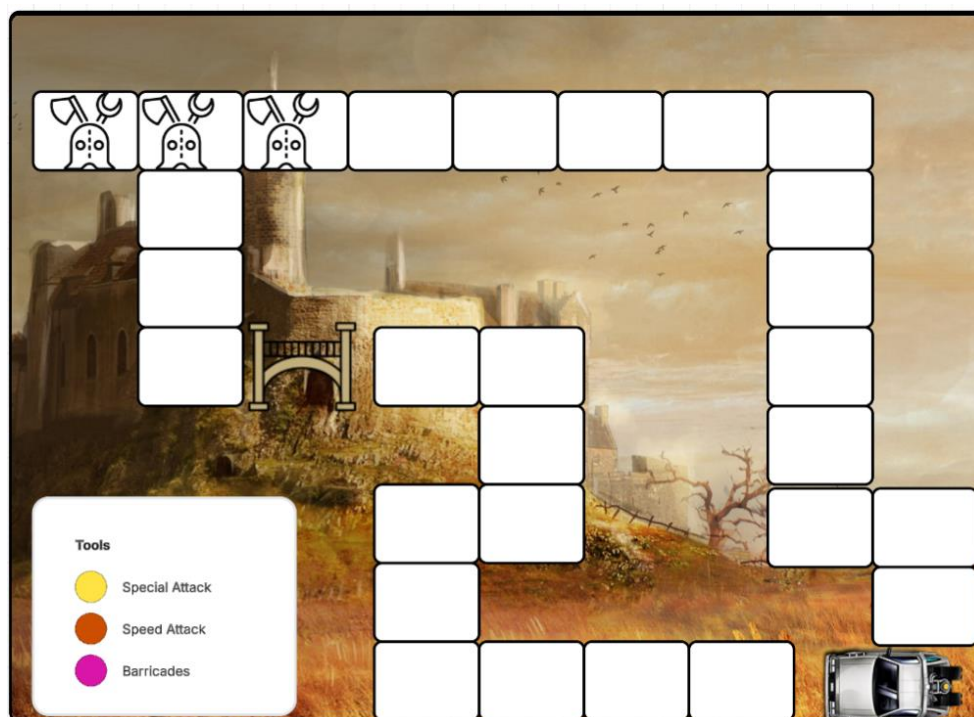


Figure 1: 1st iteration of prototype - basic map layout (wireframe style)

The second iteration of our prototypes involved mocking-up a hand-drawn version of the first level of our game, incorporating the additions and modifications we discussed as a group. We thus moved the time machine to the center of the screen and placed it inside a medieval castle “tower.” We drew paths surrounding and extending from the castle and incorporated a draw bridge to match the level theme. We also used this prototype to demonstrate the differences in the two modes of gameplay: resource-collection and traditional defense mode.



Figure 2: Hand-drawn wireframe with traditional gameplay mode



Figure 3: Hand-drawn wireframe for resource collection mode

After further group discussion, we decided to create another high-fidelity prototype with more details including an idea of the art style for the game. We thus generated the prototypes shown in Figures 4 and 5 as they provided a more realistic view of how our game might look.



Figure 4: High-fidelity wireframe with traditional gameplay mode

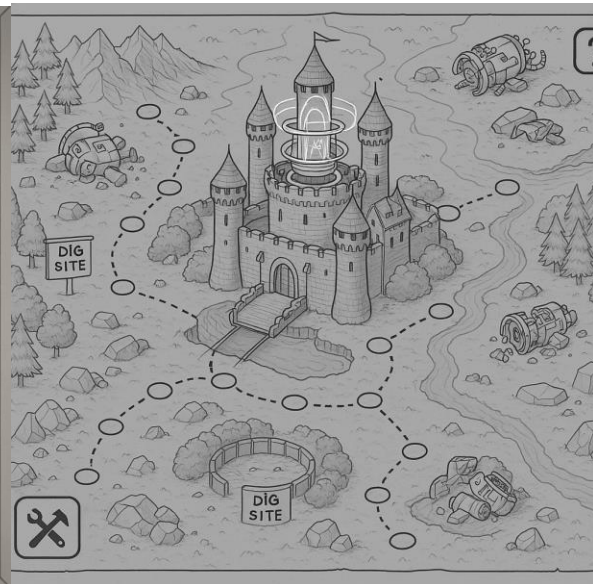


Figure 5: High-fidelity wireframe for resource collection mode

Lastly, we created prototypes in Blender and Unity to help us understand what a destructible environment might look like and how to go about developing this feature. We started a [GitHub repository](#) to store these prototypes, and the results of these efforts are shown in Figures 6 and 7 below.

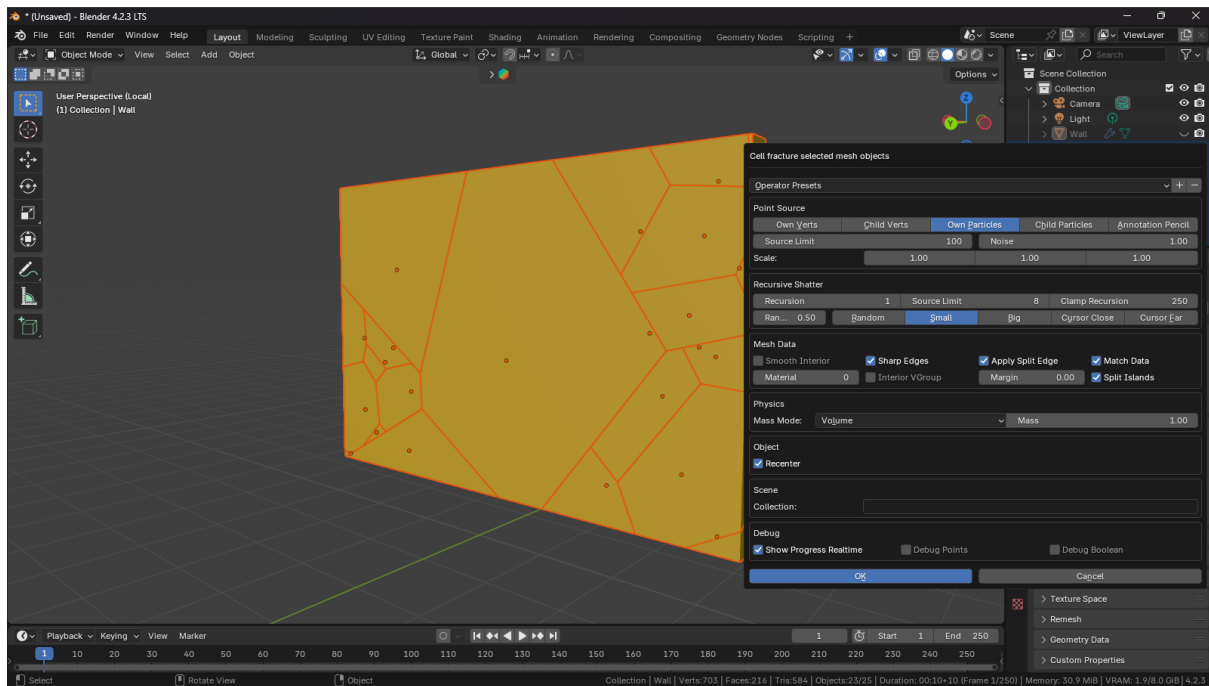


Figure 6: Prototype of a destructible barricade in Blender showing how it would break apart when hit

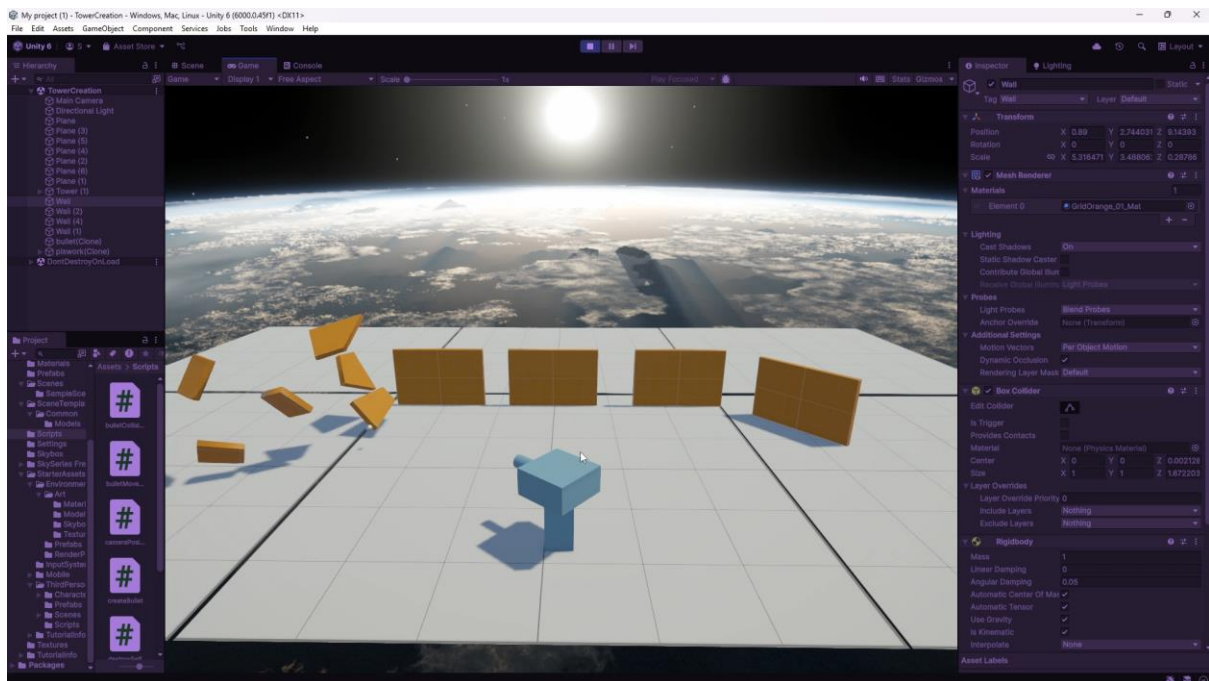


Figure 7: Prototype demonstrating the destructible barricades in a Unity environment

Key Mechanics

Note: Any features marked as optional will only be implemented if time allows or if they are critical to the core experience during development.

Movement and Interaction

- WASD for character movement; mouse for camera control and selecting interactable objects.
- E or Right-click to interact — works when next to an object (e.g. gathering resources or placing defences).

- **(Optional)** Right-click can also be used to move to a clicked location
 - **(Optional)** Support for queued actions: e.g., click multiple interaction points in sequence for efficient gathering and building.
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Two Gameplay Modes

The game switches between two primary modes, each affecting what actions the player can perform:

Mode 1: Build and Gather Mode

- This is the default state at the start of the game and between waves.
- The player controls the Traveller to gather resources and place defences such as towers, barricades, and traps.
- During this mode the player can choose to upgrade existing defences
- This mode ends when the player is ready or the pre-wave timer reaches zero.
 - **(Note:** We are still deciding whether the wave timer is fixed or player controlled to allow for more relaxed or high-pressure pacing, depending on feedback from playtesting.)

Mode 2: Attack Mode

- Triggered when a wave of enemies begins.
 - During this phase, **gathering and building are disabled**.
 - The player can use the Traveller to **fight enemies directly** using melee and ranged attacks, as well as their special abilities.
 - The player must rely on prior planning to defend against the wave while staying engaged in real time combat
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Traveller Combat Abilities

- **Basic Melee Attack**
 - Activated by pressing interact on an enemy while in close range.
 - **Low cooldown**, most used attack players will be using.
- **Ranged Attack**
 - Area-of-effect (AoE)
 - Less burst and focused damage than melee but allows for use at a distance.
 - Good for groups of enemies.
 - **Medium cooldown**, cannot be used repeatedly in quick succession to encourage more strategic use
 - **(Optional)** Add a Damage-over-time effect (DoT) to encourage players to switch between basic and ranged attacks whenever the range attack is off cooldown.
- **Buff / Utility Ability**
 - Temporarily boost player speed and reduce skill cooldowns allowing the player to attack faster.
 - **Medium cooldown**, cannot be used repeatedly in quick succession to encourage more strategic use
 - **(Optional)** Allow players to use during **Build and Gather Mode** to reduce gathering and build timers.
- **Special Attack**

- Charges up gradually over time and through repairing the time machine.
 - Deals significant damage and slows the enemy.
 - **Long cooldown**, serves as a powerful Area-of-effect (AoE) attack.
 - Useful to prevent getting overwhelmed or against a boss enemy, strategic use is essential for victory.
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Enemy Types

- **Regular Melee Enemy**
 - Low HP, short-range
 - Targets either barricades or the time machine directly when in range.
 - **Regular Ranged Enemy**
 - Low HP, ranged attacks
 - Targets either barricades or the time machine directly when in range.
 - **Boss Enemy**
 - Much higher health and stronger damage.
 - Has a “**Rage**” state after taking damage or after a timer, increasing speed and damage.
 - (**Optional**) While regular enemies ignore the player, the boss can target the hero, forcing player engagement and active defence.
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Defence Structures

- **Basic Tower**
 - Standard ranged defence.
 - Can be upgraded to improve damage and increase fire rate.
 - Requires gathered resources to be upgraded.
 - (**Optional**) **Additional Tower Types:**
 - Tower variations like area denial towers (e.g. tesla tower zapping enemies in range), slowing towers, or Area-of-effect (AoE) damage towers.
 - **Barricade**
 - Placed to block or delay enemies.
 - Limited durability and can be destroyed by all enemy types.
 - Useful for slowing down enemies.
 - **Explosive Trap**
 - Triggered by enemy proximity.
 - Deals burst damage to all enemies in range.
 - One time use, intended for strategic placement or during an emergency if getting overwhelmed.
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Destructible Environment

- **Destructible bridges:** Bridges connect enemies spawn zone and time machine along the shorter path and can be destroyed by the player to alter the enemy's pathfinding options and buy extra time for defences.
- **Barricades:** Can be placed to slow down enemies to buy time for defences, but they can also be broken down by enemies.

- These dynamic destructible environmental changes add an extra layer of strategy, allowing the player to manipulate enemy routes mid battle.
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Wave Design and Progression

- Each new wave introduces more enemies or new enemy types, increasing the challenge as the game progresses.
 - If the player neglects tower upgrades and building of defences, they risk being overwhelmed.
 - Boss enemies appear at milestone waves, forcing the player to balance all mechanics.
 - Repairing of the time machine requires gathered materials and periodical intervals
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Player Decisions:

Players must constantly choose whether to use gathered materials on repairing the time machine or on building and upgrading of defences. They'll need to decide where to place barricades to slow down enemies and when to destroy bridges to alter the enemy route and how to time their combat abilities such as saving their special attack for a boss or using it to prevent being overrun. This creates a strategic gameplay loop.

Characters and Settings

Backstory

Our protagonist, simply called the **Traveller**, is a time traveller attempting to make their way back home to both their place and time, after a long expedition through history. On their journey back however, they are attacked by fourth dimensional creatures called the **Swarm**.

The Swarm are a group of creatures capable of moving through space and time and are attracted by the Traveller's time machine. Due to the damage inflicted on the **Machine** by their attack, the Traveller is forced to make a crash landing and decides to do so in a famed medieval city containing a tower whose defences they believe they could fortify to help them weather the further expected onslaught from their enemies.

Ultimately, their end goal is to return to their time; this means ensuring their continued survival as they continue to forage for parts and resources to undo the damage to the machine dealt by the enemies.

Characters:

The player character is a time traveller stuck in a hostile time and environment making efforts to keep alive and get back home

Player Character: The Traveller

A time traveller marooned in time at a tower in a medieval city, attempting to fix their machine, while enduring the onslaught of enemy attacks.

The traveller possesses several combative abilities as already detailed in the 'Key Mechanics' section. In addition to this however, they also possess the following abilities to ensure their survival and achieve their goals:

- Fortification of the tower through set-up of defences from a central console as guided by 'The Machine'.

- Foraging abilities to assist with gathering resources with the goal of rebuilding their time machine.
- Short-term defensive shield to enable quick getaways

Side Character: The Machine (Optional)

The **Machine** can be thought of as the sentient element or artificial intelligence construct of the Traveller's damaged time machine.

The Machine, in general, helps guide the traveller's actions, always offering well-timed advice to and notifying the Traveller of important changes to their condition or environment.

The Machine essentially acts as a proxy for the game designers to provide useful information to the players to help maximise gameplay enjoyment and utility.

Enemy Characters: The Swarm

The enemies, collectively known as the Swarm, are creatures with motives mostly unknown, seemingly attempting to consume the traveller's machine (and potentially the traveller also), effectively stopping them from getting back to their home and time.

The Swarm come in three variants:

Regular Melee: Workers

These are the 'worker bees' of the swarm, their strength lying in their numbers and hive mentality. While they are generally lacking in durability and possess no direct interest in the Traveller, their actions and attempts to 'acquire' the Machine ultimately present an obstacle in the Traveller achieving his goals.

Regular Ranged: Drones

These are the 'soldier bees' of the swarm, their strength lying both in their physicality and in their more battle-hardened approach to combat. This experience informs their ability to make ranged attacks allowing them to inflict damage on a number of targets without being in the sphere of influence/attack of said targets. In this way they avoid the risk of potential damage from said targets.

For the most part however, their physicality is just for show as it mostly serves to enable their range attack abilities while offering little in the way of increased defensive capabilities over their typical 'worker' companions.

Boss: Nexus

This class of enemy represents the most difficult enemy type. Possessing much greater strength and defenses than their companions, as well as greater intelligence, they are (potentially) able to realize the secondary goal of attacking the Traveller is in line with their primary goal of 'acquiring' the Machine.

As such, they are the only enemy class to initiate direct confrontations with the Traveller character. In conjunction with their high offensive and defensive capabilities, their presence always presents a real threat to the Traveller and serves to increase the stakes of the game.

Objectives:

The game combines both short and long-term objectives in order to realize enjoyable and engaging gameplay.

The short-term objectives, which take centre stage for most of the game play, include the following:

- Ensuring the 'Traveller's survival even against the unceasing enemy waves and onslaught of attacks

- Maintaining possession and current state of the 'Machine' even as the enemy targets on it continues

The long-term goal includes:

- Enabling the Traveller to make his way back home. This requires:
 - o Gathering resources to fix the Machine as well fulfilling the prior stated short-term goals/objectives

Beginning the Game

At the start, players are immediately dropped into the game, where a brief cinematic sets the scene and introduces both the Traveller and the world. Players take direct control of the Traveller and are gradually introduced to the movement controls through an optional, unobtrusive popup. Since WASD movement and mouse controls are familiar to most players, these hints are minimal and can be ignored if already known, but remain available for those who need them.

The tutorial is seamlessly integrated into gameplay using camera pans, object highlights, and updated quest objectives, allowing players to progress naturally without being overwhelmed by menus or excessive UI. Simple visual prompts, such as "Press E to collect," appear only when near relevant objectives, keeping the interface clean and intuitive. Visual feedback, like effects shown when repairing the time machine, reinforces the impact of player actions and highlights the importance of key mechanics. New gameplay elements are introduced gradually in order of complexity, ensuring players are not overloaded with information.

Once the first wave begins and the game shifts into attack mode, players are introduced to their unique abilities one at a time, eventually gaining access to the full set. These interactions are woven into the gameplay itself; for example, gathering the first material serves as both a tutorial and a necessary step for progression, as indicated by the quest objective. To encourage experimentation and foster a sense of discovery, certain tutorials, such as those for destructible environments, are only triggered when the player finds them independently. Instead of constant direction, players are guided by narrative cues and contextual hints, allowing them to learn through exploration rather than being over-directed by tutorials.

A dedicated help icon is still available, however, the goal is for players to learn everything naturally through direct interaction in the first few minutes.

First 2-5 Minutes

The game begins with an opening cinematic showing the Traveller arriving in a hostile and strange environment and their time machine breaking down. The transition into gameplay begins with a simple gathering task which eases the player in with one of the most fundamental mechanics.

After the player finishes gathering, the objective updates, instructing the player to interact with the broken time machine. When the player interacts with it, a simple animation shows it being repaired. This interaction reinforces the importance of gathering as it introduces the long term goal of fixing the time machine to escape.

With the machine partially repaired the camera pans to a highlighted spot on the map and the objective is updated to instruct the player to build their first defence. The player uses their gathered resources to place a basic tower. This gives the player their first taste of the build mechanic and introduces resource management as resources are both used for repairing the time machine and for creating defences. The UI also shows a countdown before the first wave begins.

Here the player faces their first real decision: use the remaining materials to build more defences or continue to gather and stockpile for later all under a time constraint. While being low pressure it enforces **player agency** by allowing them to decide what they want to focus on.

As the wave begins, the player is shown the approaching enemies and the game changes into Attack Mode. The player is now unable to build and gather. Since it's an early round only a few basic melee enemies' approach, to allow the player to familiarise themselves with the various abilities of the

Traveller. They are also aided by the defences they have placed which further reinforces the importance of **strategic defence use and placement**.

Once the player has dealt with all the enemies, the game goes back to build and gather mode allowing the player to progress the repairing of the time machine and further strengthen their defences for the upcoming waves. By now the player has been introduced to the **core gameplay loop of gathering → repairing → building defences → combat → repeat**.

From here the next round begins with more pressure from new enemy variations and larger waves. This provides a challenge to the player who needs to adapt to the situation and introduces real consequences for the player's actions.

After the first few min the player has:

- Experienced both game modes and the various systems associated with each.
- Made impactful choices that affect their gameplay.
- Introduced to the core gameplay loop but given the freedom to optimise their strategy and experiment with various options.

Art Style

The game 'Tower Defence' heavily utilizes a renaissance(/medieval) themed, low polygon art style, minimizing hardware requirements while offering a clean and clear-cut look and feel.



Character:

It is currently envisioned that the character asset shall have a higher polygon count especially when compared to the city tower and environment.

The main purpose of this will be to accentuate the contrast between the character, a being displaced in time, and the environment. This should ultimately evoke in the player a sense of non-belonging and further subtly amplify player motivations to complete the game and get the character back to their home and time.

In addition to this, it is also currently intended that the character be masked to give a sense of mystery.

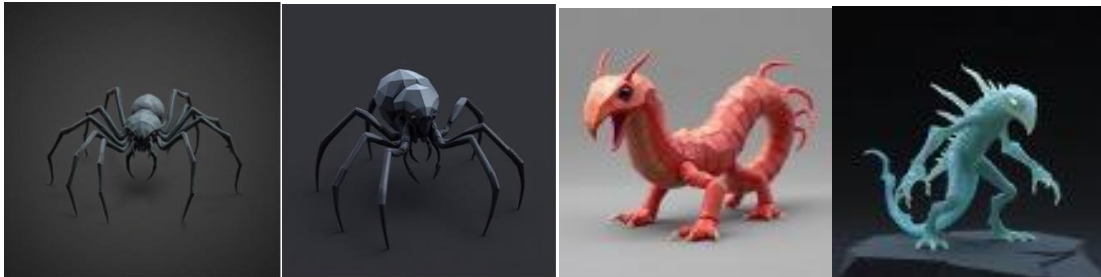


Enemies:

A similar approach as with the characters is taken to evoke a sense of non-belonging for the enemies also.

The current approach, if possible, is to utilize mid-poly assets with animated vertices to give a sense of fluidity as the enemy characters, with respect to the background story, are capable of moving through 4-D space. How far this approach is taken will depend on time constraints and available assets.

As they are enemy characters however, any relevant assets used will also need to be capable of inducing a sense of danger. As such, it is thought that the final assets to be used will combine elements of these two salient points.



Tower & City:

As previously stated, the general art style to be used will be low-polygon and renaissance themed with the motivation for the low-polygon environment contrasting with the character and enemy assets already explained.

The game itself is set in a medieval era to start with and so the art style is intended to heavily contain renaissance art elements as shown in the example below.



Also, as the tower itself is meant to be an abandoned place, the scenery and setting will be chosen to evoke a sense of isolation and abandonment. This further ties in with certain planned game mechanics like the destructible bridges mechanic also.



Audio

A number of things need to be taken into consideration with regards to the audio design of the game.

First and foremost, of priority, is maximizing gameplay immersion. For this purpose, it is intended that background music reminiscent of the medieval era will be heavily utilized. Choices for this will be heavily inspired by classic time-period dramas and movies particularly those set in medieval European cities like Old England, renaissance era Italy and the like.

In addition to this, a number of background musical pieces will also be used at appropriate moments during gameplay to heighten player tension and perceived stakes, particularly during enemy swarms etc.

In addition to background music, we also consider the audio style to be taken for sound effects. These are heavily tied to the in-game mechanics to be implemented and will be decided accordingly. They include sound effects such as 'bridge destruction', enemy damage, player damage, time-machine damage, player attack etc.

It is also intended that the UI elements of the game have their associated sounds as well. This is mostly to ensure appropriate feedback to the user during interactions with UI elements. We do not intend for these to stand out and industry suggested best UI design practices will be followed to ensure appropriate choices for these are made.

Finally, we consider any speech-related audio. These can include character voice acting, in game narration, enemy and player vocalizations etc. It is not currently intended that speech be utilized heavily in the game. If, and when, they are used, it is currently thought that it will be limited to necessity only. One area this might change might be for tutorial related or instructional narration to help guide the player in their gameplay.

SWOT Analysis

Strengths

The game's primary strength lies in its innovative combination of tower defense and resource management mechanics, which creates a unique gameplay experience that differentiates it from traditional tower defense games. The low-poly art style offers significant technical advantages, requiring less computational power and allowing the game to run smoothly on a wide range of devices, from high-end gaming computers to mobile platforms. This performance efficiency makes the game accessible to a broader audience and reduces development complexity. The time travel theme provides a compelling narrative framework that gives context to the gameplay mechanics, while the medieval setting offers familiar visual elements that players can easily understand. The dual-objective

system of defending while gathering resources adds strategic depth and prevents the repetitive feeling common in many tower defense games.

Weaknesses

One significant weakness is the challenge of balancing the two core mechanics without making either feel secondary or overwhelming. Players may find it difficult to manage both tower defense and resource gathering simultaneously, potentially leading to frustration or confusion about priorities. The time travel theme, while engaging, risks becoming convoluted if not handled carefully, as many games struggle with time travel concepts without creating plot inconsistencies. As a university project with limited development time and resources, as a team, we face constraints in creating sufficient content variety to maintain long-term player interest. The low-poly art style, while efficient, may limit the visual detail needed to create truly memorable characters and environments. Additionally, the team's inexperience in game development could lead to technical challenges or design decisions that negatively impact our final product.

Opportunities

The tower defense market is experiencing steady growth, with the mobile tower defense segment expected to reach \$1.27 billion by 2030 (Valuates Reports), representing significant commercial potential. The combination of low-poly aesthetics and strategic gameplay appeals to both casual and hardcore gamers, creating opportunities to reach diverse player demographics. The time travel theme offers potential for future expansions or sequels that could explore different historical periods or locations. The game's performance-efficient design makes it well-suited for mobile platforms, where tower defense games have shown strong success with freemium monetization models. Educational institutions and gaming communities increasingly value innovative indie projects, providing opportunities for recognition and networking within the game development industry. The medieval setting provides rich thematic material for additional content, characters, and storylines that could extend the game's appeal.

Threats

The tower defense genre is highly competitive, with numerous established titles already capturing significant market share. Larger development studios with greater resources could easily implement similar mechanics in their own games, potentially overshadowing our project. Technical challenges related to balancing real-time tower defense with resource management could prove more complex than anticipated, leading to delays or compromised gameplay. The limited development timeline of a university project restricts the ability to conduct extensive playtesting and iteration, which are crucial for tower defense games. Market saturation in the indie game space makes it increasingly difficult for new titles to gain visibility and attract players. The team's reliance on keyboard/mouse controls may limit the game's appeal compared to more modern alternatives with touch controls or gamepad support.

Production Schedule

Team member	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11
Rebekah	Kickoff, roles, setup	GDD writing	Art style research	Resource system prototype	GDD writing	Refine art assets	Feedback point	Refine art assets
Alex	Kickoff, roles, setup	GDD writing	Art style research	Resource system prototype	GDD writing	Refine systems	Holiday	Holiday

Tade	Kickoff, roles, setup	GDD writing	Core mechanics prototype	Basic art assets	GDD writing	Refine mechanics	Feedback point	Refine mechanics
Sumaylen	Kickoff, roles, setup	GDD writing	Core mechanics prototype	Enemy AI Prototype	GDD writing	Refine AI	Feedback point	Refine AI

Team member	Week 12	Week 13	Week 14	Week 15	Week 16	Week 17	Week 18	Week 19
Rebekah		Playtest #1	Feedback point	Level design	Downtime/catch up	Feedback point	Playtest #2	Finalize gameplay
Alex	Refine systems	Playtest #1	Feedback point	Tutorial implementation	Downtime/catch up	Feedback point	Holiday	Polish features
Tade		Playtest #1	Feedback point	UI/UX design	Downtime/catch up	Feedback point	Playtest #2	Polish art/UI
Sumaylen		Playtest #1	Feedback point	Sound/music	Downtime/catch up	Feedback point	Playtest #2	Bug fixing

Team member	Week 20	Week 21	Week 22
Rebekah	Finalize gameplay	Documentation submission	Drink coffee
Alex	Polish features	Documentation submission	Drink coffee
Tade	Polish art/UI	Documentation submission	Drink coffee
Sumaylen	Bug fixing	Documentation submission	Drink coffee

List of all assets

Art Assets

Name	Items/Description	Estimated Time	Priority
General (e.g. Player, Enemies, Machine & Tower)	Sprite	1 hour	Essential
	Movement animation	2 hours	Essential
	Damage effects	2 hours	Essential
	Destruction effects	2 hours	Essential
Tower	Barricades	2 hours	Essential
	Ruins	1 hour	Essential
	Greenery	1 hour	Optional
City	Houses	1 hour	Optional
	Buildings	1 hour	Nice to have
	Bridges	3 hours	Essential
UI Elements	Element text fonts	1 hour	Essential
	Health bar	1 hour	Essential
	Machine integrity status bar	1 hour	Nice to have

Audio Assets

Name	Items/Description	Estimated Time	Priority
Background Music	2-3 tracks to denote gameplay mode/status i.e. gathering or battle mode	1 hour	Essential
Sound Effects	Player movement sound	1 hour	Essential
	Enemy movement sound	1 hour	Essential
	Enemy attack sound	1 hour	Essential
	Player resource gathering sound	<1 hour	Optional
	Player resource collection sound	<1 hour	Essential
	Upgrade sound	<1 hour	Nice to have
	Player damage sound	<1 hour	Essential
	Tower defence element damage sound	<1 hour	Essential
	Tower defence element destruction sound	<1 hour	Nice to have
	Tower defence element placement sound	<1 hour	Essential
	Time machine damage sound	<1 hour	Essential
	Time machine destruction sound	<1 hour	Nice to have
UI Interaction	Menu open sound	<1 hour	Nice to have
	Menu item selection sound	<1 hour	Optional
	Game over sound	<1 hour	Nice to have
Speech	To be determined	Unknown	Optional

System Assets

Name	Description	Estimated Time	Priority
Player Controller	Handles the WASD movement and player input	2 hours	Essential
Camera Controller	Controls the third person camera and camera rotation	2 hours	Essential
Combat System	Manages the cooldowns, damage values and interactions of the various abilities in the game	24 hours	Essential
Enemy AI	Responsible for the pathfinding and behavior of the enemies	24 hours	Essential
Wave Manager	Manages wave timing and composition of enemies in the wave	12 hours	Essential
Resource Nodes	Controls the spawn location and behavior of resource nodes	8 hours	Essential
Resource Manager	Tracks the value of obtained resources and responsible for updating the associated UI elements	2 hours	Essential
Game State Manager	Controls switching between the two game modes. Also enables/disables systems per mode	4 hours	Essential
Defense Placement System	Responsible for how the placement of defenses work such as grid snapping and preview	8 hours	Essential
Defense Manager	Manages the behavior of all deployed defenses including target tracking and cooldown logic	16 hours	Essential
Upgrade System	Managing the upgrades of defense and applies stat changes	6 hours	Essential
Destructible Environment System	Controls the behavior and values of destructible objects	6 hours	Essential

UI Manager	Controls the behavior of various UI elements based on user input	10 hours	Essential
Audio Manager	Responsible for Music/SFX playback and triggers	6 hours	Essential
Tutorial System	Handles the triggers for tutorial zones and prompts	4 hours	Essential
Settings System	System for controlling audio/visual aspects based on user input	8 hours	Nice to have
VFX Controller	Responsible for particle/shader creation and control	4 hours	Nice to have
Player Stats Tracker	Logs various player stats (enemies killed, waves survived, resources collected)	2 hours	Optional
Action Queueing	Supports the ability to click several interaction points in sequence for efficient gathering/building	6 hours	Optional
Save/Load System	Creates a snapshot of game state allowing players to stop at anytime and then continue later	6 hours	Optional

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